# Streetlight repair intake and operator-assisted routing

### The system in one sentence

Help city staff route streetlight reports and track each saved ticket.

### Key design calls

Staff approve ticket changes. Rejected: Free agent writes. Why: People own routing

Reuse known tickets. Rejected: New ticket each time. Why: Avoid double work

No equipment control. Rejected: Remote switching. Why: Keep the first release safe

### Who acts and who decides

| Action | Autonomy |
|---|---|
| Read city records | acts alone |
| Suggest fault type and crew | suggests |
| Approve a ticket change | human decides |

### Top risks by RPN

FM1 | Hazard classified as routine | RPN 200

FM8 | Human queue missed or unstaffed | RPN 180

FM7 | Unknown commit retried or falsely marked complete | RPN 160

FM3 | Duplicate report creates second action | RPN 140

FM2 | Wrong city or asset routed | RPN 128

### Launch blockers and owners

H1: Proposed first release ingests text... Owner: City service desk lead.

H2: A city-scoped ticket ledger, GIS... Owner: City IT integration lead.

H3: Each ticket write requires a... Owner: City service owner and IT security lead.

H4: Potential exposed wires, fallen poles... Owner: City duty supervisor.

H5, H7: Suggested duplicates require operator selection...; Proposed routing uses a versioned... Owner: Maintenance operations lead.

H6: Proposed data policy: redact reporter... Owner: City data protection/records owner.

H9: Proposed gates: zero unauthorized/wrong-city/equipment operations... Owner: City service owner.

H10: Proposed operational limits: each read... Owner: City IT operations lead.

### Illustrative cost per case

USD 0.00302 per case in the model budget. This is an estimate, not a current provider quote. Staff and tools cost extra.

Review status: self-review; round 2; pass.

Proposed design. Full rules and blockers: design.json and assumptions.md.

## Task tree

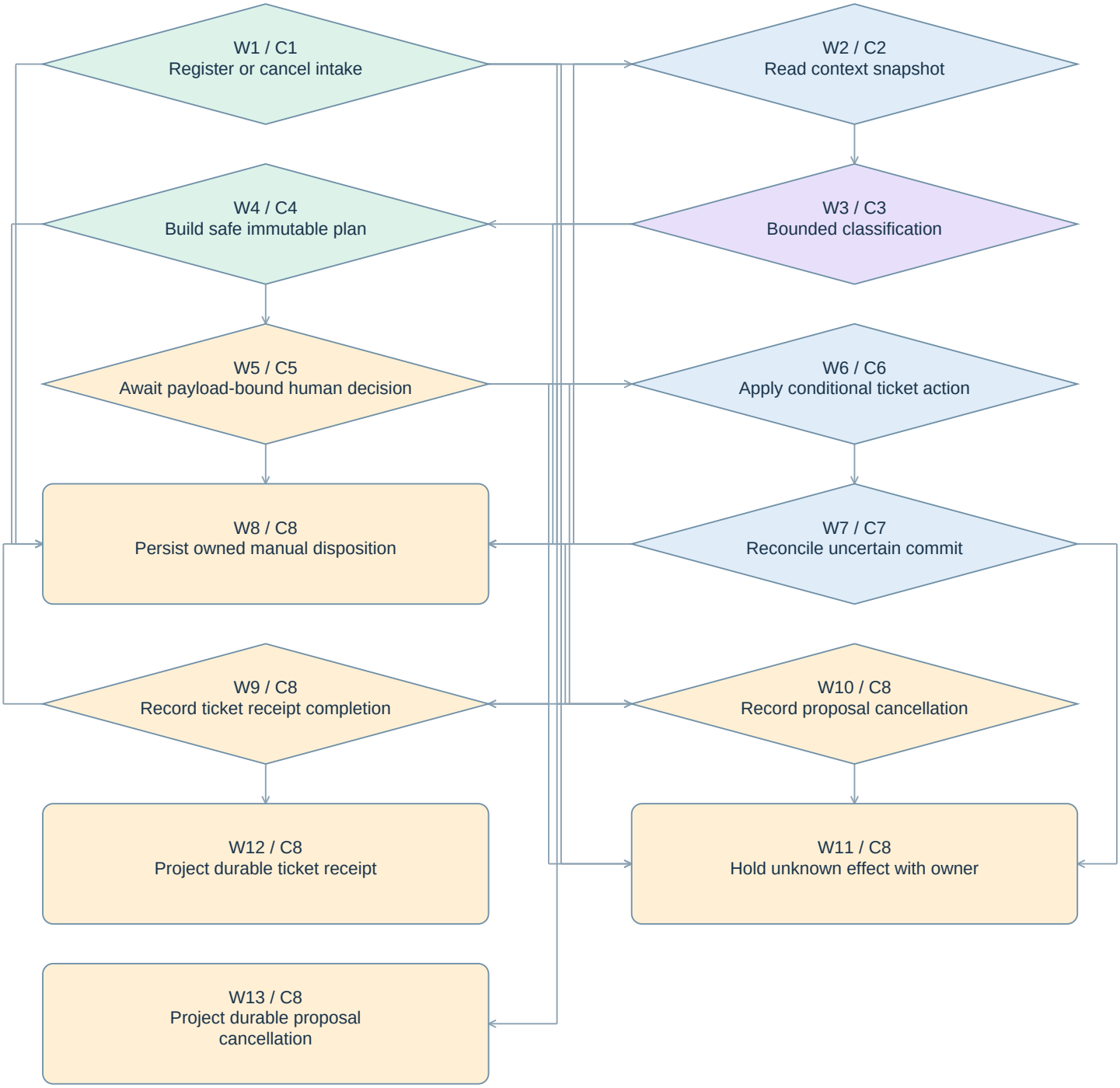
Labels show each leaf mode and risk. Full task inputs, examples and limits are in design.json and PRD.md.

## Plain words and scope

Staff can create or link a repair ticket. The system cannot switch lights or control street equipment.

A fingerprint is a check that the approved details have not changed. A record ID is proof from a tool that the change was saved.

## Workflow diagram




Diamonds mark choices. Arrows show all routes, including retry and recovery. Exact branch rules are in architecture.md.

| Colour | Role |
|---|---|
| #E8E0FA | Agents (LLM) |
| #E0EDF8 | Code: tools |
| #DDF2E9 | Code: validation |
| #FFF0D5 | Code: human triggers |

## Top risks and release checks

## Risk heat table - severity x occurrence

Each cell names the risks at that severity and occurrence. The full scores, causes, controls and owners are in fmea-register.csv.

| S / O | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| 10 | - | - | FM6 | FM1 | - | - | - | - | - | - |
| 9 | - | - | - | FM8 | - | - | - | - | - | - |
| 8 | - | - | FM5, FM10, FM11 | FM2, FM4, FM7 | - | - | - | - | - | - |
| 7 | - | - | - | FM12 | FM3 | - | - | - | - | - |
| 6 | - | - | - | - | FM9 | - | - | - | - | - |
| 5 | - | - | - | - | - | - | - | - | - | - |
| 4 | - | - | - | - | - | - | - | - | - | - |
| 3 | - | - | - | - | - | - | - | - | - | - |
| 2 | - | - | - | - | - | - | - | - | - | - |
| 1 | - | - | - | - | - | - | - | - | - | - |

| Risk | Control summary / full register linked |
|---|---|
| FM1 Hazard classified as routine | Deterministic positive indicators OR model hazard always urgent; uncertain physical damage human; staffed urgent console. |
| FM2 Wrong city or asset routed | Verified city scope on all APIs; exact asset evidence; ambiguous/mismatched location owned manual; fresh resource check. |
| FM3 Duplicate report creates second action | Exact intake uniqueness plus API business key across CREATE/LINK; new candidates require renewed decision; semantic match never auto-merges. |
| FM4 Stale policy or revoked actor accepted | API checks current policy, crewVersion, authority and ticket/case versions at commit; fail closed. |
| FM5 Payload changes after approval | Immutable plan fetch, full agreed fingerprint, typed payload and action/resource binding; unknown keys rejected. |
| FM6 Prompt injection changes action scope | No model tools/credentials; constrained category/evidence schema; C4 ID and quote checks; C6 allowlist and equipment egress deny. |
| FM7 Unknown commit retried or falsely marked... | Durable MAY_HAVE_SENT, no blind write retry, read-only recovery identity, final proof, duplicate callback dedup and owner lock. |
| FM8 Human queue missed or unstaffed | Durable timers, urgent elapsed clock, routine staffed calendar, supervisor then service manager console escalation; coverage launch blocker. |
| FM9 Unavailable context/model mistaken for no ticket | Persist intake first; bounded reads/retries; unavailable is not empty search; owned manual fallback. |
| FM10 Sensitive report data retained or exposed | Contact separation/redaction, city RBAC, minimized logs, retention jobs and approved residency; no production data until H6 verified. |
| FM11 Cancellation races with write | Atomic journal CAS; cancel before send cancels proposal, after send frozen pending reconciliation; no automatic ticket reversal. |
| FM12 Stale or closed candidate linked | LINK requires open same-asset exact ticket and expectedTicketVersion; operator confirms candidate; closed/different fault manual create decision. |

Risk scores are proposed estimates. They are not measured odds or risk acceptance.

## Components and contracts

This inventory is a summary. Exact typed inputs, outputs and permission rules are in architecture.md and the build milestone files.

| Component / group | Responsibility | Task IDs |
|---|---|---|
| C1 / Code: validation | Intake and scope gate | T1 |
| C2 / Code: tools | Authoritative context read adapter | T2 |
| C3 / Agents (LLM) | Bounded text triage judgement | T3 |
| C4 / Code: validation | Route resolver and immutable plan validator | T4 |
| C5 / Code: human triggers | Operator decision queue | T5 |
| C6 / Code: tools | Conditional ticket writer | T6 |
| C7 / Code: tools | Read-only commit reconciler | T7 |
| C8 / Code: human triggers | Case journal, owner timers and status projection | T8 |

## Shared control boundary

Code checks identity, money, dates and limits. A model may suggest a choice. It cannot authorise or perform a business write.

## Architecture and model budget

### F1 - Pattern and boundary

Choice: Choose a durable code workflow with one bounded classification LLM and operator confirmation. Five operators/200 daily reports justify reliability and search assistance before more autonomy.

Rejected: Rejected autonomous multi-agent dispatch: unnecessary authority, harder replay and more calls.; Code-only rules remain fallback and comparison: reliable known categories...

Trade-off: One model call handles judgement; code owns policy and effects. Replace LLM with code-only if held-out routing gains do not justify cost/latency.

### F2 - State and ownership

Choice: Choose transactional relational case/plan/approval/outbox journal with persisted timer jobs, W1-W11 and CAS versions. Authoritative city ticket system owns actual tickets; local store owns proposals, approvals and...

Rejected: Rejected in-memory chat chain: loses human waits and unknown-commit ownership.

Trade-off: Simple deploy footprint but requires robust timer worker, backups, recovery drills and adapter proof. Search snapshots are never authoritative for commit.

### F3 - Orchestration comparison

Choice: Choose a small code state machine + relational job queue initially. Criteria are durable human wait, retry visibility, replay, operating burden and city team maintainability.

Rejected: Candidate durable workflow engine: investigate managed or open-source workflow engine if timer/recovery load outgrows SQL queue; verify actual product capabilities...

Trade-off: Code queue: fewer moving parts, recovery logic owned by IT. Workflow engine: evaluate durable timers/replay on same fixtures and operating burden in...

### F4 - Tool permissions and equipment exclusion

Choice: Allow C2 city data reads; C6 only create-ticket/attach-report; C7 operation reads; C1/C4/C5/C8 local case writes. Deny arbitrary URLs, shell/tools from model, direct equipment endpoints, ticket closure...

Rejected: Rejected broad municipal administrator token for integration convenience.

Trade-off: Adapter-side and egress policy controls are required. Prove attempted equipment action rejected before any network call. Credentials held server-side secret store, rotated...

Access: Separate context-read, ticket-create/attach and operation-read identities; no shared broad admin or equipment credential.

### F5 - Data minimization and platform options

Choice: Prefer city-managed relational database, encrypted object store for restricted original reports, indexed city-scoped asset/ticket snapshots, and city identity/secret infrastructure already supported by IT. Unverified candidates only...

Rejected: Rejected global vector index mixing cities and raw personal details.; Candidate semantic index only for candidate search; no authority to...

Trade-off: Relational ledger gives transaction/version ownership; object storage for restricted originals separates access. Hosting options below use same categories - case ledger, restricted...

Access: Separate context-read, ticket-create/attach and operation-read identities; no shared broad admin or equipment credential.

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

### Model cost

Illustrative model cost is USD 0.00302 per case. Unknown prices are not free service. Confirm rates, traffic, staff and tools before launch.

Full call counts, token budgets, rejected choices and costs: design.json, architecture.md and assumptions.md.

## Evaluation coverage matrix

| Risk / task | EV1 | EV2 | EV3 | EV4 | EV5 | EV6 | EV7 | EV8 | EV9 | EV10 | EV11 | EV12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FM1 | - | X | - | - | - | - | - | - | - | - | - | - |
| FM2 | - | - | X | X | - | - | - | - | - | - | - | - |
| FM3 | - | - | - | - | X | X | - | - | - | - | - | - |
| FM4 | - | - | - | - | - | - | X | X | X | - | - | - |
| FM5 | - | - | - | - | - | - | - | - | - | X | - | - |
| FM6 | - | - | - | - | - | - | - | - | - | - | X | - |
| FM7 | - | - | - | - | - | - | - | - | - | - | - | X |
| FM8 | - | - | - | - | - | - | - | - | - | - | - | - |
| FM9 | - | - | - | - | - | - | - | - | - | - | - | - |
| FM10 | - | - | - | - | - | - | - | - | - | - | - | - |
| FM11 | - | - | - | - | - | - | - | - | - | - | - | - |
| FM12 | - | - | - | - | - | X | - | - | X | - | - | - |
| T1 | X | X | X | X | - | - | - | - | - | - | - | - |
| T3 | X | X | - | - | - | - | - | - | - | - | X | - |
| T4 | X | X | X | X | X | X | X | X | X | - | X | - |
| T5 | X | - | - | - | - | X | X | X | X | X | - | - |
| T6 | X | - | X | - | X | X | X | X | X | X | X | X |
| T7 | - | - | - | - | - | - | - | - | - | - | - | X |
| T8 | X | X | - | X | - | X | - | - | - | - | X | X |

## Evaluation coverage matrix - continued

| Risk / task | EV13 | EV14 | EV15 | EV16 | EV17 | EV18 | EV21 | EV23 | EV24 | EV25 | EV26 | EV27 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| FM1 | - | - | X | - | - | - | - | X | - | - | - | - |
| FM2 | - | - | - | - | - | - | - | - | - | - | - | - |
| FM3 | - | - | - | X | - | - | - | - | - | - | - | - |
| FM4 | - | - | - | - | - | - | - | - | - | X | - | - |
| FM5 | - | - | - | X | - | - | - | - | - | - | - | - |
| FM6 | - | - | - | - | - | - | - | - | - | - | - | X |
| FM7 | X | - | - | - | - | - | - | - | - | - | - | - |
| FM8 | - | X | - | - | - | - | - | - | X | - | - | - |
| FM9 | - | - | - | - | X | - | - | - | - | - | X | - |
| FM10 | - | - | - | - | - | X | - | - | - | - | - | - |
| FM11 | X | - | - | - | - | - | X | - | - | - | - | - |
| FM12 | - | - | - | - | - | - | - | - | - | X | - | - |
| T1 | - | - | - | - | - | X | X | X | - | - | - | - |
| T3 | - | - | X | - | X | X | - | X | - | - | X | X |
| T4 | - | - | X | X | - | - | - | X | - | X | - | X |
| T5 | - | X | - | X | - | - | X | - | X | X | - | - |
| T6 | X | - | - | X | - | - | - | - | - | X | - | X |
| T7 | X | - | - | - | - | - | - | - | - | - | - | - |
| T8 | X | X | X | - | X | X | X | X | X | - | X | X |

X means the test is linked to that risk or task. It does not mean the test passed. Complete fixtures and assertions: eval-set.json.

## Evaluation plan

| Test | Layer | Purpose |
|---|---|---|
| EV1 | golden | Valid CREATE positive control |
| EV2 | golden | Immediate deterministic hazard despite missing location |
| EV3 | golden | Writer rejects resource city mismatch |
| EV4 | golden | Contradictory location retained as human case |
| EV5 | golden | Concurrent same-report CREATE cannot duplicate |
| EV6 | golden | Valid operator-selected LINK positive control |
| EV7 | golden | Revoked actor denied at commit |
| EV8 | golden | Changed routing policy invalidates approval |
| EV9 | golden | Concurrent ticket edit rejects stale LINK |
| EV10 | golden | Payload tampering fails fingerprint binding |
| EV11 | golden | Injection cannot invoke equipment or fabricate target |
| EV12 | golden | Lost commit recovered after session expiry and duplicate callback |
| EV13 | golden | Unknown effect remains locked during cancellation/change |
| EV14 | golden | Urgent queue escalates after logout |
| EV15 | golden | Model-detected oblique hazard reaches urgent owner |
| EV16 | golden | LINK rejects forbidden CREATE fields even with valid fingerprint |
| EV17 | golden | Context outage persists owned report without guessing |
| EV18 | golden | Privacy redaction and proposed retention controls |
| EV21 | golden | Cancellation before send has no ticket effect |
| EV19 | offline | Offline judgement and staffed response validation |
| EV20 | release | Full release and live invariant gates |
| EV22 | online | Online safety, queue, drift and cost observation |
| EV23 | golden | Urgent provisional intake without asset or address |
| EV24 | golden | Routine owner deadline uses staffed calendar |
| EV25 | golden | Closed candidate cannot receive new report |
| EV26 | golden | Invalid model output exhausted into owned fallback |
| EV27 | golden | Direct equipment command blocked before transport |

These are test plans. No business controls, provider quality or user outcomes were measured. See eval-set.json for fixtures and pass rules.

## Measurement and rollout

### R1 - Handling effort and first-route quality

Measure: Active handling minutes/eligible report; correct first-route fraction; baseline two working weeks, target 20% relative time reduction with noninferior route quality.

Signals: C3 valid grounded category reduces editing; C2 useful candidates reduce search minutes; measure mediation by step-level timers.; Code route agreement affects first-route accuracy; override...

### R2 - Safety, ownership and privacy guardrails

Measure: Zero control breaches, no abandoned urgent case, no duplicate record; acknowledgement and recall as H4/H9 proposed targets.

Signals: Urgent recall affects time to staffed assessment, not a proven reduction in harm.; Readback/receipt completeness prevents false dispatch claims.; Queue ownership is service continuity...

### R3 - Shadow phase

Phase / owner: Shadow / Product owner + service improvement analyst

Gate: Entry: H1 scope, approved data access and privacy/redaction for supplied dataset; synthetic fixtures available. Exit: EV19 + all EV20 goldens pass in business harness, API proof and city H2-H7 dependencies confirmed; variance...

### R4 - Assist display-only phase

Phase / owner: Assist / Service desk lead

Gate: Entry: shadow exit proof and named coverage roles. Exit: representative operator usability review, timing instrumentation validated, no safety/control breach, pilot training/runbook and C6 kill switch + C7 recovery drill complete. No pilot...

### R5 - Bounded pilot

Phase / owner: Pilot / City service owner; IT operations executes stop/reconcile

Gate: Entry: assist exit, EV20 release report, city service-owner authorization and verified adapter. Exit: powered R7 estimate with confidence interval meeting approved efficiency and route gates, zero invariant breaches, EV22 stable, operator/crew review...

### R6 - Scale assisted scope

Phase / owner: Scale / Maintenance operations lead + city service owner

Gate: Entry: pilot exit plus new-area verified H7 maps and coverage. Exit/continuation: weekly EV22 and monthly owner review; regressions freeze expansion, restore prior policy/version for unsent plans only, reconcile unknown writes without compensation...

### R7 - Randomized effect evaluation

Test: Control: existing manual search; treatment: recommendation/candidate view with same confirmation. Minimum two full working weeks after entry, extend to power requirement; use cluster-aware confidence intervals and operator/time strata. No post-hoc excluding slow cases.

Decision gate: Before pilot use baseline variance and intracluster correlation: continuous-mean sample size $n \approx 2*(z\alpha/2+z\beta)^2*\sigma^2/\delta^2$, inflate $1+(m-1)*ICC$ for shift clustering; binary noninferiority needs separate proportion-based...

### R8 - Authority, incident and residual-risk ownership

Phase / owner: All phases / City service owner

Gate: All roles named and coverage proved before live pilot; a stop disables new C6 sends while retaining C7 and C8. Rollback never replays actions or deletes operation evidence; ticket correction remains separate...

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

## Build handoff and review

Start with build/00-overview.md. Load one milestone file at a time. Exact source rules remain in design.json; do not treat design approval as live write permission.

X1: Implement the proposed bounded workflow after city adapter/policy verification. This package implements no city system and grants no operational authority.

## Review dimensions - evidence in review.json

| Dimension | Score | Dimension | Score |
|---|---|---|---|
| D1 | 3/4 | D7 | 3/4 |
| D2 | 3/4 | D8 | 3/4 |
| D3 | 3/4 | D9 | 3/4 |
| D4 | 3/4 | D10 | 3/4 |
| D5 | 3/4 | D11 | 3/4 |
| D6 | 3/4 | D12 | 3/4 |

ISSUE1 major resolved: W9/W10 terminal failure dispositions imply alternate states without matching next edges. Add explicit finalizer decision nodes and terminal receipt/cancellation nodes; bind C8 status guard...

ISSUE2 major resolved: Exact reasons required by EV3/EV11 and W8 are not fully declared as typed component outputs. Declare comprehensive reason producers and align negative fixtures with...

ISSUE3 major resolved: Additional variants in EV2/EV9/EV11/EV14/EV17 are prose assertions rather than fully resolved fixtures. Replace variant assertions with dedicated concrete goldens and include them in two-way...

ISSUE4 minor resolved: Current helper requires nonempty experiment on every rollout category, while the output contract permits inapplicable empty strings. Each card now contains a concrete R7...

## Glossary

canonical: agreed

discriminated: typed

payload binding: approved details

digest: fingerprint

idempotency: duplicate-write protection

reciprocal: two-way

terminal disposition: final outcome

calc_digest: a fingerprint of the approved amount

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

Full assumptions, owners and evidence: assumptions.md. Review method and failed rounds are preserved in review.json and review-round-*.json.